

Pseudocode:

Input:

- User selects a currency from the options provided:
 - a. US\$
 - b. AU\$
 - c. Euro
- User enters a valid amount (1 to 95):
For AU\$, the amount must be a multiple of 5.
- After calculating the coin distribution, the user decides whether to continue or exit the program:
User inputs 'y' for yes or 'n' for no.

Output:

- The program outputs the number of coins needed for each denomination based on the selected currency and the input amount.
- If the user enters an invalid currency choice or amount, the program displays appropriate error messages.
- After each calculation, the program asks the user if they wish to continue or exit:
Displays the message: Do you want to continue? (y/n):

```
// High-Level Algorithm Start

Main

// Function Declarations
FUNCTION SelectCurrency
FUNCTION GetAmountInput(currencyChoice)
FUNCTION SetCurrencyDenominations(currencyChoice,
denominations)
FUNCTION CalculateCoins(amountInput, denominations, coinCount,
size)
FUNCTION DisplayCoinResults(coinCount, denominations, size)
FUNCTION CheckContinueOrExit

Initialize continueRunning to true

DECLARE Size as 4 // Size for the number of denominations
WHILE continueRunning IS true
    selectedCurrency = SelectCurrency()
    inputAmount = GetAmountInput(selectedCurrency)

    int denominations[Size]; // Array for denominations
    SetCurrencyDenominations(selectedCurrency, denominations);

    int coinCount[Size]; // Declare coin count array
    FOR i from 0 to Size - 1
        coinCount[i] = 0; // Initialize all elements to 0
    END FOR

    CalculateCoins(inputAmount, denominations, coinCount,
Size);

    DisplayCoinResults(coinCount, denominations, Size);

    continueRunning = CheckContinueOrExit()
```

```
    END WHILE
    RETURN 0
END Main
```

```
// High-Level Algorithm End
```

```
// Low-Level Algorithms Start
```

```
Step 1 Refinement
```

```
FUNCTION SelectCurrency
    DISPLAY "Select a currency:"
    DISPLAY "1) US$"
    DISPLAY "2) AU$"
    DISPLAY "3) Euro"

    INPUT currencyChoice
    WHILE currencyChoice is less than 1 OR greater than 3
        DISPLAY "Invalid choice. Please select 1, 2, or 3."
        INPUT currencyChoice
    END WHILE
    RETURN currencyChoice
END FUNCTION
```

```
Step 2 Refinement
```

```
FUNCTION GetAmountInput(currencyChoice)
    DISPLAY "Enter an amount (1 to 95):"
    INPUT amountInput

    IF currencyChoice equals 2 (AU$) THEN
        WHILE amountInput < 1 OR amountInput > 95 OR amountInput
is not divisible by 5
            DISPLAY "Invalid amount. Must be a multiple of 5
within 1 to 95."
            INPUT amountInput
        END WHILE
    ELSE
        WHILE amountInput < 1 OR amountInput > 95
            DISPLAY "Invalid amount. Enter a value from 1 to 95."
            INPUT amountInput
        END WHILE
    END IF

    RETURN amountInput
END FUNCTION
```

```
Step 3 Refinement
```

```
FUNCTION SetCurrencyDenominations(currencyChoice, denominations)
    SWITCH currencyChoice
        CASE 1 (US$):
            denominations = [50, 25, 10, 1]
        CASE 2 (AU$):
            denominations = [50, 20, 10, 5]
```

```

        CASE 3 (Euro):
            denominations = [20, 10, 5, 1]
        END SWITCH
    END FUNCTION

```

Step 4 Refinement

```

FUNCTION CalculateCoins(amountInput, denominations, coinCount,
size)
    FOR i from 0 to size - 1
        coinCount[i] = amountInput divided by denominations[i]
        amountInput = amountInput modulo denominations[i]
    END FOR
END FUNCTION

```

Step 5 Refinement

```

FUNCTION DisplayCoinResults(coinCount, denominations, size)
    DISPLAY "Coin distribution:"
    FOR i from 0 to size - 1
        IF coinCount[i] > 0
            DISPLAY coinCount[i] + " x " + denominations[i] + "
cent coins"
        END IF
    END FOR
END FUNCTION

```

Step 6 Refinement

```

FUNCTION CheckContinueOrExit
    DISPLAY "Do you want to continue? (y/n):"
    INPUT userResponse
    WHILE userResponse is not 'y' AND userResponse is not 'n'
        DISPLAY "Invalid response. Enter 'y' or 'n'."
        INPUT userResponse
    END WHILE
    IF userResponse equals 'y' THEN
        RETURN true
    ELSE
        RETURN false
    END IF
END FUNCTION

```

```

// Low level Algorithm End

```